

TrES-1b

R-1450

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_140602 · created 2026-08-01T09:26:29+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2456810.7729 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.143 +/- 0.02
Transit depth (Rp/Rs)^2	2.05 +/- 0.56 [%]
Orbital Inclination (inc)	89.83 +/- 0.58
Airmass coefficient 1 (a1)	0.908 +/- 0.01
Airmass coefficient 2 (a2)	0.092 +/- 0.031
Scatter in the residuals of the lightcurve fit is	1.11 %
Best Comparison Star	#2 - [83, 51]
Optimal Aperture	3.51
Optimal Annulus	9.44
Transit Duration (day)	0.0613 +/- 0.0015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.143 $\pm$ 0.02 vs 0.1358 (deviation 0.29 σ)
Duration fit vs published	0.0613 d vs 0.1042 d (deviation 22.9 σ)
Mid-time O-C	2.8 min
Depth SNR	5.7
Residual scatter	1.11 %

Light curve

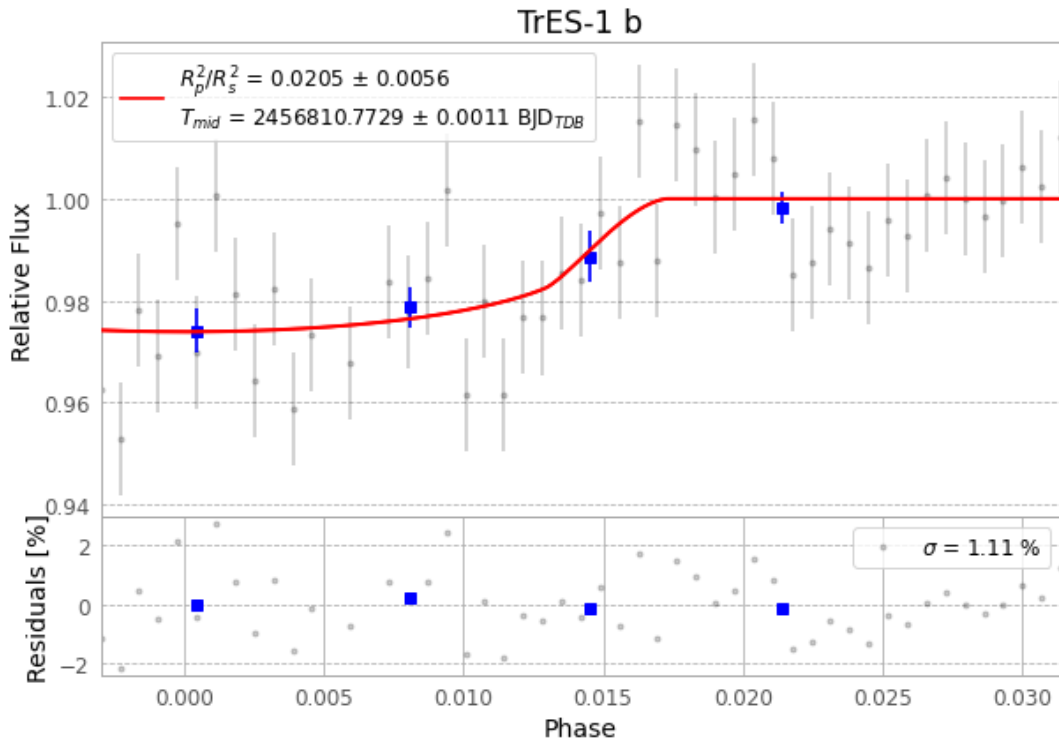


Figure 1 — FinalLightCurve_TrES-1 b_2014-06-01.png (SHA-256 3b9b35638c1a0bf6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [173, 245], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b32e1c4fa693b199...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

51 FITS frames (33 MB), TRES-1140602061514.FITS → TRES-1140602084513.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-140602.log (4cb61ffadad3...), FinalLightCurve_TrES-1 b_2014-06-01.png (3b9b35638c1a...), FinalLightCurve_TrES-1 b_2014-06-01.csv (cabf0bec6f46...)

Compute resources

Wall time 150.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 09:26Z.